Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/29/2005
PI Name:	Glazier, James Alexander Ph.D.		
Project Title:	Comparative Study Of Morphogenesis In 1g And Simulated Microgravity/Subproject B		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	47405-7105	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	03/01/2003	**End Date:**	02/28/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1619		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of this project is to compare fundamental embryological morphogenetic phenomena under normal and zero gravity conditions. Specific areas to be considered include gastrulation, limb development and cell sorting. Tasks include development of biomechanical models for devlopment and experiments under normal and simulated microgravity conditions. These experiments and simulations will be coordinated with supporting and parallel research by Dr. Rusty Lansford (CalTech) and Dr. Gabor Forgacs (U. Missouri, Columbia). Task Objective: Develop protocols necessary for assays of effects of simulated microgravity on cell motility and signaling during developmental processes. Task Significance: Since many simulated microgravity effects on development are quite subtle, it is essential to have a complete range of quantitative techniques to assess changes in cell behavior related to cytoskeletal disruption under microgravity.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have discovered a fundamentally new patterning mechanism operating in chick limb beyond classical positional coding and reaction-diffusion. This mechanism has fundamentaly different character and is probably important in a large variety of other developmental processes in other animals. Understanding the mechanisms of development is a key step in developing therapies to treat developmental abnormalities, design regenerative medical therapies and treat degenerative diseases in adults. The microfluidics cell, when further developed will assist in automating and streamlining drug discover and evaluation and may have applications in detecting biotoxins.
Task Progress:	We have developed cadherin-GFP fusion protein transformed cell lines for a variety of cadherins and expression levels and show that variations in expression of a single cadherin suffice to induce cell sorting under normal gravity, thus supporting the Differential Adhesion Hypothesis. We have installed a Rotating Chamber Cell Culture System and are beginning work on analysis of cell aggregatesunder simulated microgravity.
Bibliography Type:	Description: (Last Updated: 12/29/2020)
Articles in Peer-reviewed Journals	Merks RHM, Glazier JA. "A cell-centered approach to developmental biology." Physica A 2005 Jul;352(1):113-30. http://dx.doi.org/10.1016/j.physa.2004.12.028 , Jul-2005
Articles in Peer-reviewed Journals	Cickovski TM, Huang C, Chaturvedi R, Glimm T, Hentschel HG, Alber MS, Glazier JA, Newman SA, Izaguirre JA. "A framework for three-dimensional simulation of morphogenesis." IEEE/ACM Trans Comput Biol Bioinform. 2005 Oct-Dec;2(4):273-88. PMID: 17044166 , Oct-2005
Articles in Peer-reviewed Journals	Chaturvedi R, Huang C, Kazmierczak B, Schneider T, Izaguirre JA, Glimm T, Hentschel HG, Glazier JA, Newman SA, Alber MS. "On multiscale approaches to three-dimensional modeling of morphogenesis." J R Soc Interface. 2005 Jun 22;2(3):237-53. PMID: 16849182 , Jun-2005
Articles in Peer-reviewed Journals	Lambert J, Cantat I, Delannay R, Renault A, Graner F, Glazier JA, Veretennikov V, Cloetens P. "Extraction of relevant physical parameters from 3D images of foams obtained by X-ray tomography." Colloids and Surfaces A: Physicochemical and Engineering Aspects 2005 Aug;263(1-3):295-302. http://dx.doi.org/10.1016/j.colsurfa.2005.01.002 , Aug-2005
Articles in Peer-reviewed Journals	Dan D, Mueller C, Chen K, Glazier JA. "Solving the advection-diffusion equations in biological contexts using the cellular Potts model." Phys Rev E Stat Nonlin Soft Matter Phys. 2005 Oct;72(4 Pt 1):041909. PMID: 16383422 , Oct-2005
Dissertations and Theses	Wei Z. "LIVING MATTER: GENERIC MECHANISMS IN EMBRYONIC TISSUE MORPHOGENESIS." Indiana University, Department of Physics, May 2005. , May-2005
Dissertations and Theses	Cheng C. "Dynamics of Cell Movement and Tissue Motion in Gastrulation and Micromass Cell Culture." Indiana University, Department of Physics, January 2005. , Jan-2005
Dissertations and Theses	Wong S-K. "A Cursory Study of the Thermodynamics and Mechanical Properties of Monte-Carlo Simulations of the Ising Model" The University of Notre Dame, Department of Physics, May 2005. , May-2005